DREChanNet: Patient-Relative iEEG Model for Channel-Wise Epileptogenic Zone Localization

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Abstract. Accurate localization of the epileptogenic zone (EZ) is essential for presurgical evaluation of patients with drug-resistant focal epilepsy. Intracranial electroencephalography (iEEG) provides direct recordings of seizure-related brain activity, yet identifying epileptogenic contacts remains challenging because background electrophysiological activity and electrode coverage vary substantially across patients. These differences complicate the interpretation of absolute signal measurements and highlight the need to account for patient-specific recording context. We propose a patient-relative (PR) plug-in that expresses physiological features or learned representations relative to the distribution of recorded channels within each patient. This label-free operation can be incorporated as an input transformation or a lightweight correction to an existing predictor. We further introduce DREChanNet, a compact ictal localization model integrating six physiological descriptors with learned raw-signal representations. Same-channel pre-onset contrasts characterize seizure-related changes, cross-seizure aggregation preserves response magnitude and variability, and PR contextualizes both branches before prediction. We investigate within-patient physiological associations and held-out-channel reference prediction, followed by baseline comparisons, component ablations, cross-source transfer, seizure-sufficiency analysis, and channel ranking. Evaluation includes an ictal cohort of 80 patients and a patient-based interictal cohort of 124 patients. PR improves macro-F1 by an average of 9.03 percentage points across eight ictal baselines and improves the separately fitted interictal readout. These findings support patient-relative contextualization as a reusable approach to channel-level localization and establish DREChanNet as an integrated model for ictal iEEG analysis.

Keywords: drug-resistant epilepsy; epileptogenic zone localization; intracranial electroencephalography; multicenter generalization; patient-relative learning

1 Introduction

Drug-resistant epilepsy (DRE) creates a localization problem with direct consequences for treatment. Surgery can provide seizure freedom for appropriately selected patients after medication failure [1–3], and international recommendations support timely surgical evaluation [4]. The central decision is which tissue to treat while preserving neurological function (Fig. 1). Intracranial electroencephalography (iEEG) informs this decision through measurements at implanted contacts [5,6].

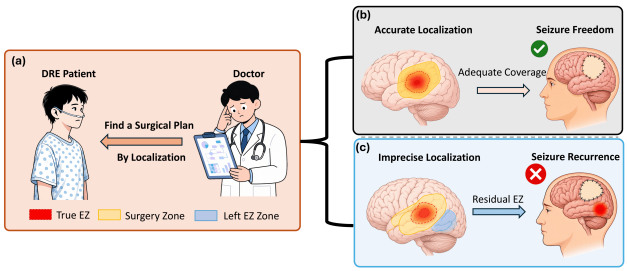


Figure 1: Clinical motivation for EZ localization. (a) Presurgical assessment identifies a treatment target. (b) Coverage of the seizure-generating tissue supports seizure freedom. (c) Residual epileptogenic tissue can sustain seizures. The surgical region and the underlying epileptogenic target need not have identical boundaries.

Localization connects a tissue-level treatment target to selectively sampled electrophysiology. The epileptogenic zone (EZ) denotes tissue whose treatment is required for seizure freedom; iEEG measures activity at locations selected to test a presurgical hypothesis [5,7]. Coverage therefore reflects clinical suspicion and implantation constraints. Human microelectrode recordings further show that large ictal field potentials can extend beyond the territory of intense neuronal recruitment [8]. The seizure onset zone (SOZ), early propagation, and treatment volume describe different stages of localization. A recent international survey documented differing clinical definitions of the SOZ, while a Delphi study established confidence grades for EZ evidence [9,10]. A channel score therefore belongs to a clinical inference task conditioned on observed brain regions, recording state, and clinical annotations.

Quantitative approaches extract complementary localization information. Ictal indices combine fast activity with recruitment timing [11]; interictal approaches characterize waveform events, spectral changes, and network dynamics [12–14]; and neural models learn representations from the recorded signals [15,16]. Recent syntheses identify spatial sampling, label definitions, and validation across patients as central challenges in translating such measurements into clinical decisions [17–19]. Together, these studies motivate a question that cuts across the choice of biomarker or encoder: against which reference should a new patient's channel be interpreted?

The reference matters because physiological variation is structured. An atlas of normal iEEG demonstrates substantial regional differences in spectral activity [20], and anatomical normative mapping uses this variation to interpret patient abnormalities [21,22]. Recording time also affects the stability of channel abnormality maps [23]. Thus, a channel value carries information about its biological and recording context as well as its relationship to the clinical target. Cross-patient learning must preserve useful within-patient differences while accommodating the baseline distributions associated with each new implantation. Alongside an external population reference, another source of context is already available when a patient is assessed: the distribution of that patient's observable channels. It indicates whether a coordinate is relatively elevated or reduced within the implanted set, without requiring the channel labels at inference.

We make this patient-relative reference an explicit, reusable part of localization. The proposed patient-relative (PR) plug-in standardizes each feature or embedding coor-

dinate across the patient's channels. In a linear readout, this transformation produces patient-dependent coordinate weights and an offset, allowing the same learned predictor to interpret different patients through their own reference statistics. PR can replace the physiological feature inputs of an existing classifier or supply a learned correction to a frozen model's output. This separates the reference operation from the architecture used to encode the signal and makes its contribution testable across models.

We further develop DREChanNet, a compact ictal model that combines physiological and raw-signal evidence. Six physiological descriptors provide interpretable measurements, while a temporal encoder captures waveform structure. Same-channel pre-onset subtraction represents temporal change, and cross-seizure means and standard deviations retain recurrent responses and their variability. PR processing then expresses both branches relative to the patient's channel set before a shared head combines them. The model compares a channel with its earlier activity and with other channels in the patient.

The evaluation follows this sequence of reasoning. Within-patient feature associations establish the relevance of the descriptors; held-out-channel distribution prediction tests the reference itself; and paired baseline comparisons and component ablations assess its contribution to localization. Patient-held-out evaluation on the 80-patient ictal cohort examines source transfer, seizure availability, and ranking. A 124-patient interictal evaluation derived from Omni-iEEG[24] tests PR with an existing backbone. The contributions are:

- A PR plug-in that makes the observed patient channel distribution available to feature classifiers and learned representations, supported by feature-association and reference-prediction analyses.
- DREChanNet, a 9,329-parameter ictal model integrating physiological and temporal representations through pre-onset referencing, cross-seizure summaries, and PR processing.
- A transparent, reproducible evaluation pipeline integrating the most comprehensive accessible ictal and interictal cohorts under strict patient-based definitions, establishing a verifiable benchmark across classification, ranking, source transfer, and seizure availability.

2 Related Work

Quantitative iEEG localization combines signal and network evidence. The epileptogenicity index links fast activity to the timing of ictal recruitment[11]. Human recordings distinguish recruited seizure cores from surrounding penumbral territories[8], with sustained ictal high-frequency activity informing that distinction[25]. Interictal approaches combine epileptiform discharges, oscillatory events, and cross-frequency coupling[12], with useful localization information also demonstrated in routinely sampled frequencies below 45 Hz[26]. Dynamical perturbations, directed connectivity, and network suppression characterize channel participation in the recorded system[13,14,27,28]. Together, these studies motivate complementary physiological and temporal representations, whose contributions we examine through association and ablation analyses.

Reference distributions and learned encoders offer complementary ways to organize this evidence. The normal iEEG atlas identifies region-dependent spectral structure[20], and normative mapping relates deviations in activity or connectivity to epileptogenic tissue[21,22]. Their reference distributions are indexed by anatomy. Temporal analyses assess their stability across recording segments[23]. In parallel, SEEG-Net uses multiscale convolutions and domain-generalization training for cross-subject pathological activity detection[15], and multiscale neural models have localized seizure-onset channels from brief interictal recordings[16]. SEEGformer combines signal representations, anatomical information, and cross-channel attention in an individual-specific detection and localization model[29]. PR makes a different reference directly available to these representations: the distribution of observable channels within the patient being assessed. This common operation can support joint training or an added readout.

Clinical interpretation depends on how methods are evaluated and how their targets are defined. Reviews of quantitative iEEG and computational DRE research identify opportunities for clinical use alongside challenges in spatial sampling, label construction, and validation[17–19]. An ILAE systematic review of connectivity-based outcome prediction further motivates standardized reporting and patient-level data sharing[30]. Channel-label agreement, outcome association, and prospective clinical benefit provide distinct forms of evidence. Shared resources support their investigation: iEEG-BIDS standardizes recording organization[31], while Omni-iEEG harmonizes recordings and metadata for multiple epilepsy tasks[24]. Within these resources, the prediction unit remains consequential because windows and channels are nested within patients; evaluation must match the intended use on new patients[32]. Our patient-held-out evaluation links reference quality to classification, ablation, and ranking, with explicit cohort construction and patient-equal summaries.

3 Methods

3.1 Task, Cohorts, and Patient-Level Evaluation

For patient p, let $\mathscr{C}_p$ be the set of observable implanted channels and $y_{pc} \in \{0,1\}$ the clinical annotation of channel c, with EZ encoded as 1. The task is to assign an EZ score q_{pc} and a binary decision to each channel. Our approach first characterizes physiological changes around seizure onset, then expresses channel representations relative to the patient's own channel set (Fig. 2). DREChanNet combines this patient-relative (PR) information in physiological and raw-signal branches; the same PR operator can also be attached to an existing predictor.

The ictal cohort includes 80 postoperative seizure-free patients, 256 seizures, and 7,635 channels, including 1,743 annotated EZ channels, from four data sources (Table 1). EZ/NEZ denotes source-specific clinical localization annotations, used as clinical target surrogates. The study received approval from the Second Hospital of Lanzhou University (2023A-765) and the Children's Hospital of Fudan University (2024-396).

Five outer folds contain 16, 16, 17, 15, and 16 patients. Each development partition is separated into fitting (FIT) and validation patients, with 13 validation patients per fold. All channels and recordings of a patient remain together. FIT

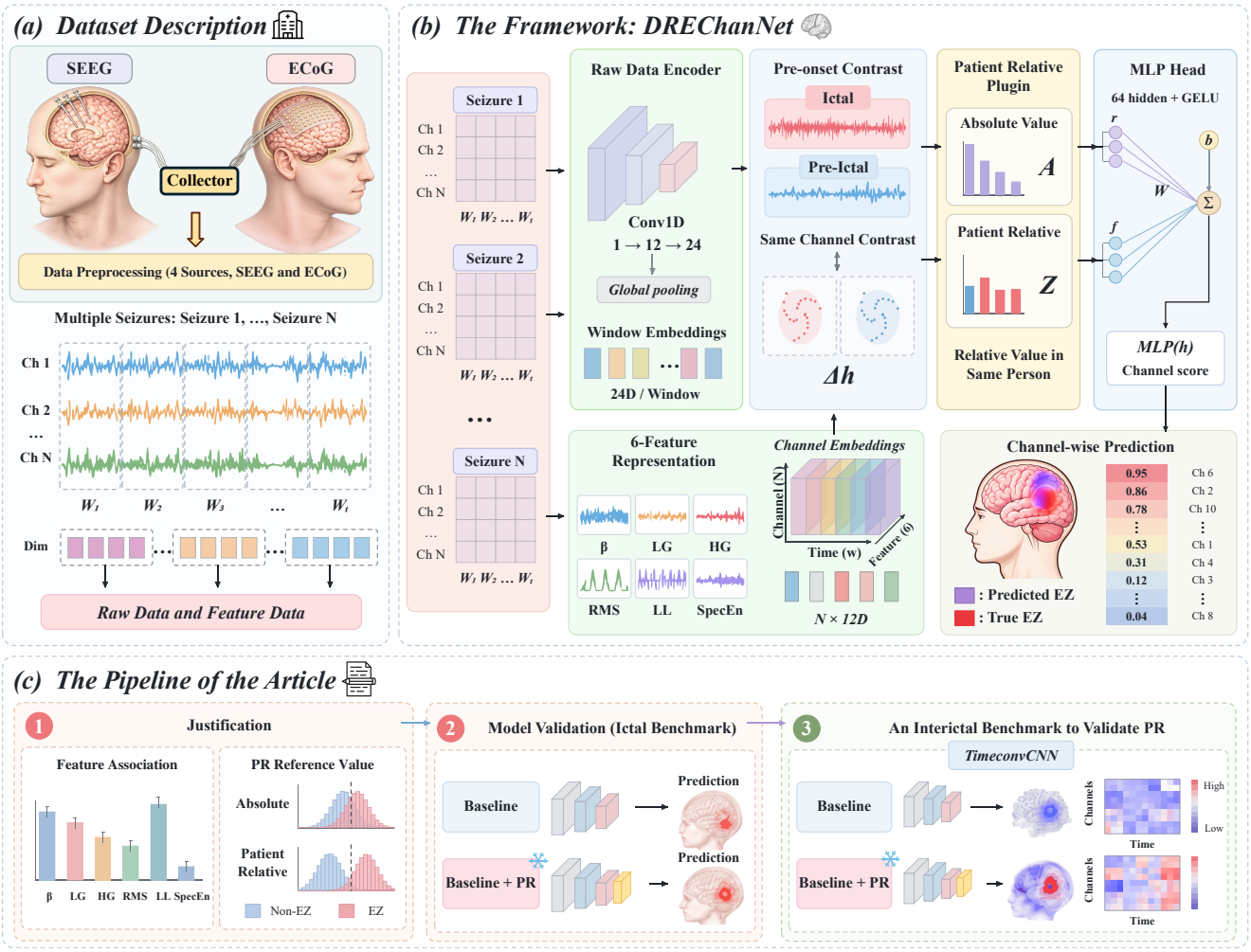


Figure 2: DREChanNet combines physiological summaries and raw temporal representations. Each branch uses a patient-relative representation before a shared prediction head. Same-channel pre-onset subtraction defines temporal changes, whereas PR defines each channel's position within the patient's channel distribution. The exact aggregation and fusion are given in Eqs. (9)–(11).

patients determine learned parameters and population preprocessing, whereas validation patients determine checkpoints and decision thresholds. Outer patients are used for evaluation. Their observable unlabeled channels supply their own PR reference at inference.

3.2 Physiological Channel Representations

Each valid 2-s window sampled at 250 Hz provides six measurements: beta power (13–30 Hz), low-gamma power (30–80 Hz), partial high-gamma power (80–125 Hz), root-mean-square amplitude (RMS), line length (LL), and spectral entropy (SE; 10–125 Hz). With N = 500, window duration T = 2 s, and Welch spectrum S(f), they are computed as

$$
\begin{aligned}
P_{[a,b]} &= \log\left(1+\int_a^b S(f)\,df\right),\\
\mathrm{RMS} &= \sqrt{N^{-1}\sum_{n=1}^{N} x_n^2 + 10^{-8}},\\
\mathrm{LL} &= T^{-1}\sum_{n=2}^{N} |x_n - x_{n-1}|,\\
\mathrm{SE} &= -\frac{\sum_{m=1}^{M}\pi_m \log \pi_m}{\log M}, \qquad \pi_m = \frac{S(f_m)}{\sum_{r=1}^{M} S(f_r)}.
\end{aligned}
\tag{1}
$$

Welch estimation uses 500 samples and 0.5-Hz bins, numerical integration uses the trapezoidal rule, and the high-gamma band includes the available Nyquist bin. Entropy uses the M bins between 10 and 125 Hz, with 0 log 0 = 0.

Table 1: Ictal localization cohorts. EZ/NEZ percentages refer to valid implanted channels.

Cohort	Patients	Seizures	Channels	EZ/NEZ (%)
HUP [33]	36	117	3,667	16.8/83.2
Multi-site [34]	15	45	925	17.7/82.3
LZU	21	72	2,141	38.2/61.8
Fudan	8	22	902	16.2/83.8
Total	**80**	**256**	**7,635**	**22.8/77.2**

Let $\psi_j(x_{pcsw})$ denote measurement j in window w of seizure record s. We first compute a same-channel pre-onset reference over valid pre-onset windows $\mathscr{B}_{pcs}$ and then average the resulting changes over all valid windows $\mathscr{W}_{pcs}$ in that seizure record:

$$
\begin{aligned}
b_{pcs,j} &= \frac{1}{|\mathscr{B}_{pcs}|}\sum_{w\in\mathscr{B}_{pcs}} \psi_j(x_{pcsw}),\\
d_{pcs,j} &= \frac{1}{|\mathscr{W}_{pcs}|}\sum_{w\in\mathscr{W}_{pcs}} [\psi_j(x_{pcsw}) - b_{pcs,j}].
\end{aligned}
\tag{2}
$$

For the S_{pc} available seizures, the mean and population SD are

$$m_{pc,j} = \frac{1}{S_{pc}} \sum_s d_{pcs,j},$$
$$v_{pc,j} = \sqrt{\frac{1}{S_{pc}} \sum_s (d_{pcs,j} - m_{pc,j})^2}. \quad (3)$$

Concatenating $(m_{pc,j}, v_{pc,j})$ for the six measurements yields the 12-dimensional physiological vector f_{pc}. Cross-seizure SD equals zero for the seven patients with one seizure. The temporal reference in Eq. (2) compares a channel with its own pre-onset activity; the PR reference below compares different channels within a patient.

We assess within-patient EZ association using the signed rank-biserial effect

$$D_{pj} = \frac{2U_{pj}}{n_{p,\mathrm{EZ}} n_{p,\mathrm{NEZ}}} - 1, \quad (4)$$

where U is the Mann–Whitney statistic, with half credit for ties between EZ and NEZ values. Positive D indicates larger values in EZ channels. Patient effects receive equal weight; 10,000 patient-bootstrap resamples provide nominal 95% intervals. Two-sided Wilcoxon signed-rank tests use Benjamini–Hochberg correction across the 12 coordinates. Source-specific means and proportions of positive effects describe their distribution across cohorts.

3.3 The Patient-Relative Operator

For any channel representation $z_{pc} \in \mathbb{R}^d$, PR computes coordinate-wise moments over the patient's available channels:

$$\mu_{pj} = \frac{1}{C_p} \sum_{c \in \mathscr{C}_p} z_{pcj},$$
$$\sigma^2_{pj} = \frac{1}{C_p} \sum_{c \in \mathscr{C}_p} (z_{pcj} - \mu_{pj})^2, \quad (5)$$

$$\mathscr{R}_p(z_{pcj}) = \frac{z_{pcj} - \mu_{pj}}{\sigma_{pj}}. \quad (6)$$

Equation (6) applies to supported, nonconstant coordinates. It specifies where a channel lies in the patient's observed distribution using the observable channel set without its clinical labels. For a linear predictor, its effect is explicit. Writing $D_p = \mathrm{diag}(\sigma_p)$ on supported coordinates,

$$w^{\mathsf{T}} \mathscr{R}_p(z) + b = (D_p^{-1} w)^{\mathsf{T}} z + b - w^{\mathsf{T}} D_p^{-1} \mu_p. \quad (7)$$

A shared predictor therefore acquires patient-dependent coordinate weights and an offset. In multiple dimensions, the changed coordinate weights can alter within-patient channel order; the operator is not limited to shifting a decision threshold. For the transformation in Eq. (6), positive affine recoding of a supported coordinate leaves its standardized value unchanged.

For physiological features, a coordinate is supported when its patient SD exceeds 10^{-8} times its FIT-derived population scale; otherwise it is assigned zero. Moments use available finite values. After patient standardization, missing values are imputed and coordinates are standardized using an imputer and scaler fitted on FIT patients only. Thus the order is patient-relative transformation followed by FIT-derived preprocessing, and PR replaces the original 12 coordinates rather than doubling their dimension. Raw-representation PR computes moments in double precision and replaces a denominator at or below 10^{-8} with 1; constant coordinates map to zero.

To test reference transfer independently of classification, each patient undergoes 100 fixed, label-independent half-channel partitions. One half supplies the patient moments and the other is assessed, then the roles are exchanged. The comparison uses identical partitions and training-standardized evaluation coordinates. Global and same-source references use training-channel moments. Location-only uses the reference-half patient mean with global SD; scale-only uses global mean with reference-half patient SD; full PR uses both patient moments. Each defines a marginal Gaussian forecast for an unchanged held-out value x, scored by the continuous ranked probability score (CRPS)[35]:

$$\mathrm{CRPS}(\mu, \sigma; x) = \sigma \left[u(2\Phi(u) - 1) + 2\phi(u) - \pi^{-1/2} \right], \quad (8)$$

where $u = (x - \mu)/\sigma$ and Φ, ϕ are the standard normal distribution and density. At $\sigma = 0$, CRPS equals $|x - \mu|$. We average coordinates, held-out channels, and repetitions within patients before averaging patients equally. Each of the 80 patients is assessed once under the reference experiment's fixed allocation; global and source moments use its corresponding training patients. Source-stratified paired patient bootstrap with 10,000 resamples provides nominal 95% intervals.

3.4 DREChanNet for Ictal Localization

The physiological branch applies feature PR to f_{pc}. The raw branch encodes valid windows with a shared two-layer temporal CNN: 12 and 24 output channels, kernel sizes 9 and 7, stride 4, GELU after each convolution, and global temporal average pooling. Before encoding, each waveform is centered, divided by its own population SD with a 10^{-6} floor, and clipped to [-8, 8]. The resulting embedding h_{pcsw} has 24 coordinates.

For each channel and seizure, we subtract its mean pre-onset embedding:

$$\Delta h_{pcsw} = h_{pcsw} - \frac{1}{|\mathscr{B}_{pcs}|} \sum_{u \in \mathscr{B}_{pcs}} h_{pcsu}. \quad (9)$$

Separate averages over the onset-relative stages $I_1 = [0, 8)$ s and $I_2 = [8, 20)$ s retain early and later ictal activity:

$$a_{pcsk} = \frac{\sum_w m_{pcsw} \mathbf{1}[t_w \in I_k] \Delta h_{pcsw}}{\sum_w m_{pcsw} \mathbf{1}[t_w \in I_k]}, \quad k \in \{1, 2\}, \quad (10)$$

where m is the validity mask and the expression is defined for nonempty stages. Across available seizures, each stage contributes its mean and population SD, producing a $2 \times 2 \times 24 = 96$-dimensional raw vector r_{pc}. Empty stages do not contribute to pooling. Raw PR transforms this vector across the patient's channels.

Let $\widetilde{r}_{pc} \in \mathbb{R}^{96}$ denote the processed raw vector and $\widetilde{f}_{pc} \in \mathbb{R}^{14}$ the processed physiological vector, whose 12 active coordinates occupy a fixed-width implementation with two zero coordinates. A shared 64-unit GELU head produces the NEZ logit:

$$h^{\mathrm{joint}}_{pc} = \mathrm{GELU}(\alpha_r W_r \widetilde{r}_{pc} + \alpha_f W_f \widetilde{f}_{pc} + b),$$
$$\ell_{pc} = w^{\mathsf{T}} h^{\mathrm{joint}}_{pc} + b_o, \quad q_{pc} = \mathrm{sigmoid}(-\ell_{pc}), \quad (11)$$

with fixed dimension-based scales $\alpha_r = \sqrt{110/192}$ and $\alpha_f = \sqrt{110/28}$. The two branches therefore contribute additively before the shared nonlinearity. DREChanNet contains 9,329 trainable parameters. Branch-only variants retain the corresponding input block and head; component ablations remove feature PR, raw PR, both PR operations, or the raw pre-onset reference; the physiological DELTA construction is retained in that ablation.

3.5 Learning and Integration With Existing Models

Training assigns equal weight to patients using one complete patient per optimizer update. For $\rho = N_{\mathrm{EZ}}^{\mathrm{FIT}}/N_{\mathrm{NEZ}}^{\mathrm{FIT}}$, the patient loss is

$$\mathscr{L}_p^{\mathrm{BCE}} = -\frac{1}{C_p}\sum_c [y_{pc}\log q_{pc} + \rho(1-y_{pc})\log(1-q_{pc})]. \tag{12}$$

For class $k \in \{\mathrm{EZ}, \mathrm{NEZ}\}$, let t_{pc}^k be its binary target and s_{pc}^k its predicted probability: $(t^{\mathrm{EZ}}, s^{\mathrm{EZ}}) = (y, q)$ and $(t^{\mathrm{NEZ}}, s^{\mathrm{NEZ}}) = (1-y, 1-q)$. Its soft F1 is

$$F_{p,k}^{\mathrm{soft}} = \frac{2\sum_c t_{pc}^k s_{pc}^k}{\max(\sum_c t_{pc}^k + \sum_c s_{pc}^k, 10^{-8})}. \tag{13}$$

The combined objective is

$$\mathscr{L}_p = 0.75\mathscr{L}_p^{\mathrm{BCE}} + 0.25\left[1 - \frac{F_{p,\mathrm{EZ}}^{\mathrm{soft}} + F_{p,\mathrm{NEZ}}^{\mathrm{soft}}}{2}\right]. \tag{14}$$

The mean of $\mathscr{L}_p$ across FIT patients is the training objective. AdamW uses learning rate 3×10^{-4} and weight decay 10^{-3}, with gradient clipping at 1. Training has a 30-epoch budget, a six-epoch minimum, and six-epoch stopping patience. Validation selects the checkpoint and a threshold on the NEZ-probability grid 0, 0.005, . . . , 1, prioritizing patient-mean macro-F1 and then EZ-F1. A channel is called EZ when 1-q_{pc} is below that threshold. Feature baselines comprise linear SVM, random forest, XGBoost, and an MLP. They use the same 12 coordinates, replaced by their patient-standardized values in the PR condition. Required imputers and population scalers are fitted on FIT patients. Raw baselines are channel-level adaptations of TimeConv-CNN[24], EEGNet[36], SEEG-Net[15], and SEEGformer[29] to the common 2-s window protocol. Their channel embeddings are formed by averaging windows within seizures and retaining cross-seizure mean and SD. Their frozen original EZ logit a_{pc}^0 receives a learned linear PR correction:

$$a_{pc}^{\mathrm{PR}} = a_{pc}^0 + \beta^{\mathsf{T}}\mathscr{R}_p(h_{pc}) + b. \tag{15}$$

Zero initialization recovers the original predictor. Only the correction and its validation threshold are fitted; the encoder and original classifier remain fixed. The correction uses patient-equal weighted BCE and AdamW with learning rate and weight decay 10^{-3}. The TimeConv seizure-sufficiency analysis retains its original end-to-end PR readout, as identified in its table.

3.6 PR Adaptation to Interictal Recordings

We construct a patient-based evaluation from Omni-iEEG[24], comprising 124 patients, 137 interictal records, and 10,202 labeled canonical patient–channel pairs (Table 2). Repeated observations of a patient's channel are combined into one evaluation unit. Good, nonresected channels from seizure-free patients are labeled normal; otherwise, good SOZ channels are labeled pathological. Remaining good channels provide unlabeled reference context. Patients with both labeled classes are included. These eligibility rules and channel consolidation define a different evaluation set from the published Task 2 cohort. The original benchmark uses a subject-level 60%/40% split; we use five patient-held-out folds with equal-patient endpoints.

Table 2: Patient-based interictal subset of Omni-iEEG after channel filtering and deduplication.

Characteristic	Evaluation cohort
Patients	124
Interictal recordings	137 EDF files
Unique patient–channel pairs	10,202
Pathological / normal	1,113 / 9,089
Sources (patients)	Open-iEEG (103), HUP (10), SourceSink (11)
Evaluation	Five patient-held-out folds

Each 60-s recording at 1 kHz is divided into 30 nonoverlapping 2-s segments. The frozen TimeConv backbone used for this evaluation yields 32-dimensional embeddings and logits. Repeated records are averaged within each segment, followed by averaging across segments. PR uses all observable good channels, including unlabeled context, with denominator $\sqrt{\mathrm{Var}(h_p) + 10^{-5}}$. The pathological-channel logit is

$$a_{pc}^{\mathrm{PR}} = a_{pc}^0 + \beta^{\mathsf{T}}\widetilde{h}_{pc} + b_{g(p)}, \tag{16}$$

where g(p) identifies one of three sources. The 32 coefficients and three source biases add 35 zero-initialized parameters. Patient-wise AdamW uses learning rate 10^{-3}, weight decay 10^{-4}, and pathological-class weight 2, with an additional 10^{-3} quadratic penalty on the coefficients and source-frequency-weighted biases.

Within each outer FIT population, threefold patient-level cross-validation selects the readout epoch from 50 epochs using AUROC and then AP. The selected epoch's inner out-of-fold readout predictions determine its classification threshold, after which the readout is fitted on all outer FIT patients. This inner validation concerns the readout; the frozen backbone was trained on outer FIT patients. The baseline threshold uses its outer FIT predictions. Outer patients are excluded from backbone and readout fitting.

3.7 Evaluation Measures and Additional Analyses

Classification and ranking metrics are calculated within each patient and then averaged equally across patients. Repeated-run summaries first average each patient's endpoint across runs. We report accuracy, AUROC, average precision, macro-F1, class-specific F1, mean reciprocal rank (MRR), and Top-1 hit rate. Ranking follows decreasing EZ scores. If $E_p(K)$ is the number of annotated EZ channels among the first K channels, precision is $E_p(K)$/K, recall is $E_p(K)/n_{p,\mathrm{EZ}}$, and hit rate is $\mathbf{1}[E_p(K) > 0]$. Reciprocal rank is the inverse rank of the first annotated EZ channel.

Source-held-out evaluation excludes an entire source from training and reports the displayed four-run descriptive average, the equal-source mean, and the lowest source mean. Seizure sufficiency uses the same 57 patients with at least three seizures and 5,735 common channels, comparing one,

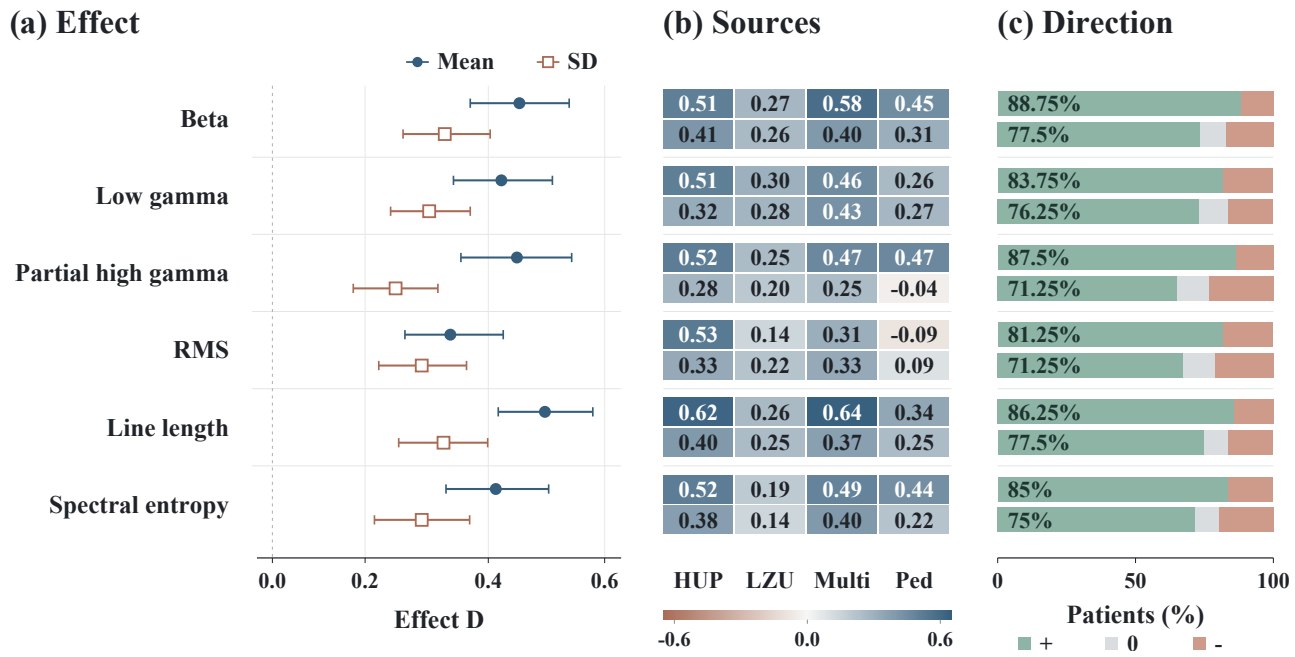


Figure 3: Within-patient associations between physiological descriptors and EZ annotations. (a) Patient-mean signed rank-biserial effects, D, with nominal 95% patient-bootstrap confidence intervals. Paired symbols denote cross-seizure means and standard deviations (SDs). (b) Mean effects in each data source. (c) Fractions of patients with positive, zero, or negative effects; labels give the positive fraction. All 80 patients are included.

two, and all available seizures. All seizure subsets are evaluated when at most 50 exist; otherwise 50 fixed subsets are sampled. Checkpoints and thresholds remain frozen, and subset endpoints are averaged within patients. The Top-K figure uses one frozen run per model and nominal 95% source-stratified patient-bootstrap intervals.

3.8 Statistical Analysis

The analysis uses patient-equal summaries. The ictal cohort includes 80 patients and the interictal cohort includes 124 patients; the seizure-sufficiency analysis includes 57 patients. Feature association uses two-sided Wilcoxon signed-rank tests with Benjamini–Hochberg correction across 12 coordinates. Nominal 95% patient-bootstrap intervals are used for feature associations, and source-stratified paired patient-bootstrap intervals are used for the reference comparison; each uses 10,000 resamples. Data preprocessing, metrics, and additional resampling procedures are specified above.

4 Results

Table 3: Prediction of unseen channels from label-free reference channels. CRPS is evaluated in shared training-standardized coordinates; intervals are nominal 95% patient-bootstrap confidence intervals.

Method	CRPS ↓	95% CI
Global Reference	0.4509	[0.3793, 0.5373]
Same-Source Reference	0.4526	[0.3783, 0.5414]
Location-only	0.3132	[0.2858, 0.3480]
Scale-only	0.4357	[0.3507, 0.5372]
Full PR	**0.1734**	**[0.1315, 0.2267]**

4.1 Physiological Evidence Within Patients

The six descriptors provide a physiological basis for channel comparison (Fig. 3). All 12 coordinates have positive overall signed rank-biserial effects, and the mean coordinate of each descriptor has a larger effect than its corresponding SD. Positive effects occur in 81.25–88.75% of patients for the mean coordinates and 71.25–77.50% for the SD coordinates. Beta-power mean is positively associated with EZ labels in 88.75% of patients, followed by partial high-gamma-power mean in 87.50% and line-length mean in 86.25%. The source-wise panel shows how the magnitudes vary across cohorts. Together, the panels motivate comparing physiological summaries within each patient's implanted set.

4.2 Predictive Value of the Patient Reference

Patient-specific reference statistics predict the distribution of held-out channels more accurately than the population references in Table 3. Full PR achieves a CRPS of 0.1734, compared with 0.4509 for the global reference, a 61.5% reduction. The Location-only and Scale-only references yield 0.3132 and 0.4357, respectively; using both patient statistics gives the lowest error. The same-source reference scores 0.4526. Because the channels used to estimate each patient reference are separated from the channels being scored, this result supports the reference's predictive value beyond the channels from which it was estimated.

4.3 PR Across Baselines and DREChanNet

Adding PR increases macro-F1 in all eight baseline comparisons in Table 4. The mean gains across these paired comparisons are 9.03 percentage points in macro-F1, 10.49 points in EZ-F1, and 6.58 points in AUROC. Linear SVM improves from 46.06% to 62.47% macro-F1, while the frozen TimeConv-CNN with a PR residual improves from 49.11% to 54.87%. These comparisons demonstrate the utility of patient-relative inputs in feature classifiers and patient-relative corrections in learned representations.

DREChanNet combines these two forms of evidence. It reaches 64.38% macro-F1, 76.67% AUROC, 57.08% average precision, 78.77% MRR, and 69.58% Top-1 accuracy. Its

Table 4: Comparison on the 80-patient localization cohort using five patient-held-out outer folds. Results are patient-equal means (%).

Methods	Input	ACC↑	AUROC↑	AP↑	Macro-F1↑	EZ-F1↑	NEZ-F1↑	MRR↑	Top-1↑
Original Baselines									
Linear SVM	12D	64.19	**68.65**	**48.16**	46.06	28.73	63.40	64.83	53.75
Random Forest	12D	62.95	60.95	42.76	48.71	**31.32**	66.09	61.74	50.00
XGBoost	12D	67.67	63.02	44.91	**51.71**	30.79	72.63	64.05	52.50
MLP	12D	65.65	66.11	47.63	49.67	30.01	69.32	**68.39**	**58.75**
TimeConv-CNN	Raw	66.76	62.06	43.15	49.11	26.70	71.53	57.01	43.75
EEGNet	Raw	67.94	56.01	34.52	44.17	12.57	75.77	47.50	30.00
SEEG-Net	Raw	**72.85**	45.06	26.38	42.22	3.08	**81.37**	34.10	22.50
SEEGformer	Raw	62.59	53.45	31.88	42.45	15.96	68.95	44.38	26.25
Baselines + Patient-Relative Methods									
Linear SVM + PR	12D	72.28	**74.36**	**54.32**	**62.47**	**45.21**	79.72	**73.88**	**65.00**
Random Forest + PR	12D	**74.90**	71.14	51.00	59.72	37.46	**81.99**	68.29	60.00
XGBoost + PR	12D	74.45	71.42	51.54	59.26	37.02	81.49	73.44	63.75
MLP + PR	12D	73.13	71.53	51.19	60.21	39.86	80.56	70.56	57.50
TimeConv-CNN + PR	Raw	67.47	64.68	44.58	54.87	35.91	73.83	60.69	45.00
EEGNet + PR	Raw	69.38	58.93	35.99	50.10	20.66	79.53	52.46	38.75
SEEG-Net + PR	Raw	67.68	60.18	37.12	51.29	24.21	78.36	54.15	38.75
SEEGformer + PR	Raw	64.19	55.76	33.98	48.46	22.72	74.20	50.30	32.50
Average Improvement		+4.11	+6.58	+5.04	+9.03	+10.49	+7.58	+7.72	+7.97
Our Method									
DREChanNet-12D	12D	75.08	74.96	54.20	62.49	42.46	82.51	74.49	64.58
DREChanNet-Raw	Raw	75.70	72.88	53.59	61.47	39.26	**83.68**	76.23	67.92
DREChanNet	Full	**76.05**	**76.67**	**57.08**	**64.38**	**45.16**	83.59	**78.77**	**69.58**
Improvement vs Baseline		+3.20	+8.01	+8.92	+12.67	+13.84	+2.22	+10.38	+10.83

macro-F1 is 12.67 percentage points above the strongest original baseline and 1.91 points above the strongest PR-augmented baseline. The feature-only and raw-only variants reach 62.49% and 61.47% macro-F1, respectively, compared with 64.38% for their joint model.

4.4 Contributions of the Model Components

Table 5: Component ablation of DREChanNet.

Configuration	F1	AUC	AP	EZ-F1	F1
Full	**64.38**	**76.67**	**57.08**	**45.16**	**69.58**
w/o Feature PR	62.15	73.65	54.38	40.71	68.75
w/o Raw PR	62.72	76.36	56.17	43.90	66.25
w/o Both PR	49.26	74.97	55.41	37.08	67.08
w/o Pre-onset	61.70	73.17	53.58	41.03	67.92
Raw-only	61.47	72.88	53.59	39.26	67.92
Feature-only	62.49	74.96	54.20	42.46	64.58

The ablation results connect the joint model's performance to its reference operations (Table 5). Removing feature PR changes macro-F1 from 64.38% to 62.15%, and removing raw PR changes it to 62.72%. Removing both gives 49.26%, corresponding to a 15.11-percentage-point difference computed from the unrounded scores. Each branch alone also yields lower macro-F1 than the joint model. The raw branch's same-channel pre-onset contrast contributes a further distinction: without it, macro-F1 is 61.70% and AUROC is 73.17%, compared with 64.38% and 76.67% for the full model. The component comparisons therefore support combining physiological and learned temporal representations with both temporal and patient-level references.

4.5 Source Transfer and Seizure Availability

In the source-held-out evaluation, DREChanNet attains the highest equal-source mean and worst-source value for both reported metrics (Table 6). The corresponding macro-F1 values are 0.6199 and 0.5649, and the AUROC values are 0.7400 and 0.6652. Full-model macro-F1 ranges from 0.5649 to 0.6498 across the four held-out sources. The joint model also exceeds either individual branch in the equal-source summary. These descriptive results assess transfer across HUP, LZU, the public multicenter source, and the pediatric Fudan source.

The seizure-sufficiency analysis holds the evaluated patients, channels, checkpoints, and thresholds constant while varying the number of seizures available at inference (Table 7). For DREChanNet with PR, macro-F1 increases from 60.94% with one seizure to 62.26% with two and 63.34% with all available seizures. EZ-F1 increases from 38.62% to 41.41% and 43.48%, respectively. The corresponding macro-F1 values without PR are 50.06%, 50.40%, and 49.54%. Thus, a single seizure already supports localization, while additional seizures improve the reported model's balanced classification and recovery of annotated EZ channels.

4.6 Channel Prioritization and an Illustrative PR Case

In the frozen-run ranking analysis, DREChanNet with PR places at least one EZ channel among the first one, three, and five channels in 67.50%, 87.50%, and 90.00% of patients, respectively (Fig. 4). At five reviewed channels, patient-mean precision is 63.00% and EZ recall is 21.98%. The corresponding PR results are 56.50% precision and 19.70% recall for SVM, and 46.75% and 15.18% for TimeConv-CNN.

Table 6: Exploratory leave-one-source-out localization across four sources, summarized over four reported repetitions. Center mean weights the sources equally; worst is the lowest source mean. Scores are proportions.

		Held-out Source				Summary	
Metric	**Method**	**HUP**	**LZU**	**Multicenter**	**Pediatric**	**Center Mean**	**Worst**
Macro-F1	TimeConv-CNN	0.4066	0.4701	0.5000	0.3217	0.4246	0.3217
	SVM + PR	**0.6623**	0.5240	0.6148	0.5950	0.5990	0.5240
	Raw-only	0.6098	0.5376	0.6316	0.6111	0.5975	0.5376
	Feature-only	0.6583	0.5633	0.5988	0.5862	0.6017	0.5633
	DREChanNet (Full)	0.6498	**0.5649**	**0.6482**	**0.6167**	**0.6199**	**0.5649**
AUROC	TimeConv-CNN	0.5341	0.5752	0.7235	0.5216	0.5886	0.5216
	SVM + PR	0.7866	0.6266	**0.8228**	0.6575	0.7234	0.6266
	Raw-only	0.7434	0.6273	0.7607	0.6951	0.7066	0.6273
	Feature-only	**0.7946**	0.6482	0.8220	0.6727	0.7344	0.6482
	DREChanNet (Full)	0.7813	**0.6652**	0.8071	**0.7062**	**0.7400**	**0.6652**

Table 7: Seizure sufficiency using frozen models on the same 57 patients and 5,735 common channels. Scores are percentages. TimeConv uses its original end-to-end PR readout in this analysis.

Model	Seizures	PR	Macro-F1	AUROC	EZ-F1
SVM	1	No	47.29	69.11	19.18
		Yes	60.64	71.90	41.36
	2	No	47.63	68.69	26.44
		Yes	61.42	73.43	43.51
	All	No	46.13	69.79	27.48
		Yes	62.08	74.74	45.32
TimeConv	1	No	48.35	60.81	19.86
		Yes	49.18	63.15	17.37
	2	No	49.49	60.81	27.18
		Yes	49.95	58.92	22.12
	All	No	50.27	61.67	30.56
		Yes	49.85	58.86	21.54
DREChanNet	1	No	50.06	71.02	27.26
		Yes	60.94	71.50	38.62
	2	No	50.40	74.95	34.04
		Yes	62.26	74.08	41.41
	All	No	49.54	76.55	35.76
		Yes	63.34	75.65	43.48

These measures quantify the concentration of annotated EZ channels in a short review list.

Figure 5 provides a channel-level illustration of the PR comparison using the separately identified development encoder. In this case, the number of correctly detected EZ channels increases from 2 of 10 to 5 of 10, with one NEZ channel assigned a positive prediction after PR. The fixed channel order makes the changed decisions directly visible alongside the clinical annotations.

4.7 Application to Interictal Recordings

Table 8: Patient-equal interictal performance on Omni-iEEG under patient-held-out fivefold evaluation.

Metric	Baseline	+PR	Improvement
AUROC ↑	0.7622	**0.8118**	**+4.97 pp**
AP ↑	0.4773	**0.5235**	**+4.63 pp**
Macro-F1 ↑	0.6104	**0.6558**	**+4.55 pp**
Pathological-F1 ↑	0.3135	**0.3923**	**+7.88 pp**
MRR ↑	0.6075	**0.6590**	**+5.15 pp**
Top-1 ↑	0.5081	**0.5645**	**+5.65 pp**

The patient-based Omni-iEEG evaluation uses the cohort

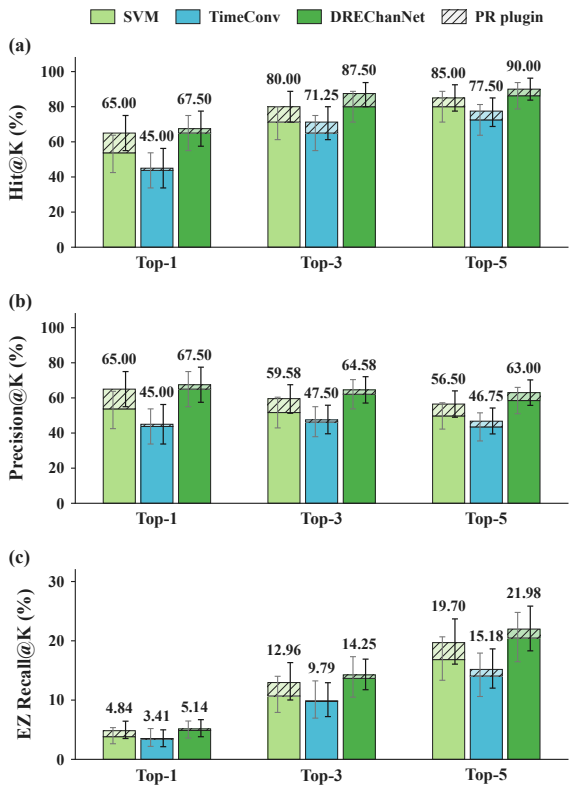


Figure 4: Channel prioritization for SVM, TimeConv-CNN, and DREChanNet. (a) Hit rate, (b) precision, and (c) EZ recall at the first one, three, and five channels. Solid bars show the baseline and hatched extensions the PR increment; labels give the PR result. Error bars are nominal 95% source-stratified patient-bootstrap intervals. Each model uses one frozen run, whereas Table 4 averages the reported DREChanNet runs.

specified in Table 2: 124 patients, 137 recordings, and 10,202 unique labeled patient–channel pairs. With a frozen TimeConv backbone, adding the PR readout improves all six reported endpoints (Table 8). AUROC increases from 0.7622 to 0.8118, average precision from 0.4773 to 0.5235, and macro-F1 from 0.6104 to 0.6558. Pathological-channel F1 increases from 0.3135 to 0.3923; MRR increases from 0.6075 to 0.6590, and Top-1 accuracy from 0.5081 to 0.5645. These results extend the evaluated application of the PR plug-in to interictal representations, without requiring the ictal architecture or its pre-onset input.

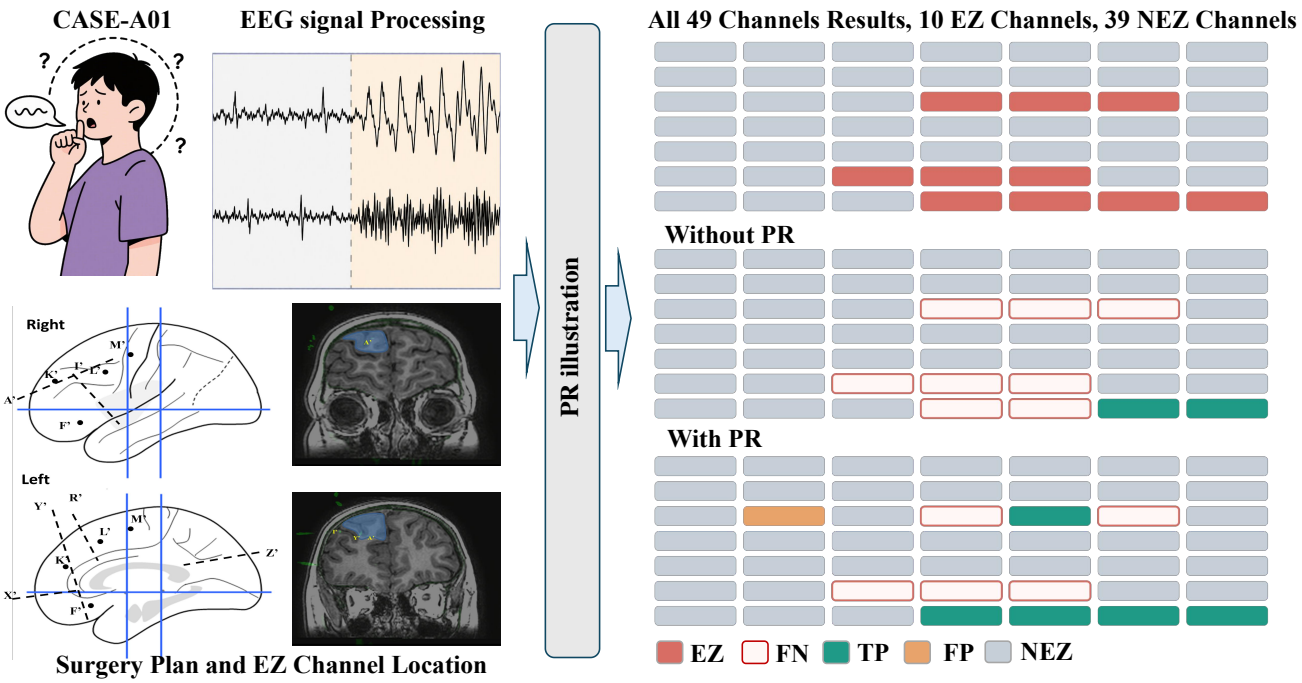


Figure 5: Illustrative PR comparison for CASE-A01: 49 channels, including 10 annotated EZ and 39 NEZ channels. Recording and surgical-planning panels illustrate the workflow.

5 Discussion

Computational localization faces a coupled problem: the clinical target is inferred, the brain is sampled selectively, and the meaning of a measured channel depends on the patient in whom it was recorded. Recent reviews identify these relationships among the main obstacles to translating quantitative iEEG into surgical practice [17–19]. Our study connects them through a defined sequence of evidence: physiological association, prediction of a patient's channel distribution, PR-based localization, and evaluation across patients, recording conditions, and seizure states.

5.1 Clinical Targets and the Meaning of Ground Truth

Table 9: Clinical observations used to assess localization and their interpretation.

Observation	Meaning for validation
Clinical SOZ	Interpreted onset within sampled contacts; its definition may include early propagation [9].
Resection or ablation	The intervention actually delivered; agreement measures overlap with treatment, with contact-specific necessity still undetermined [37].
Seizure outcome	Patient-level response at a defined follow-up; supports confidence in localization without specifying a unique channel map [10].
Algorithmic abnormality	A signal-derived deviation or score; its interpretation depends on the physiological measurement and reference distribution [17,21].

The EZ is inferred from several observations that describe different biological and clinical objects. Its surgical definition concerns tissue whose treatment is necessary for seizure freedom, whereas the SEEG tradition also emphasizes the spatial and temporal organization of the electroclinical seizure [5,38,37]. These distinctions remain consequential in current practice. In a survey of 321 experts from 39 countries, 55.8% defined the SOZ by onset alone, while 42.1% included early propagation [9]. A binary channel label can therefore encode different clinical definitions even when datasets use the same name.

Resection introduces another distinction. Seizure freedom after treating a region supports the adequacy of that intervention at the patient level. It does not identify which individual contacts within the treated region were necessary for the result. Anatomical constraints and the planned procedure can produce a treatment region extending beyond the tissue essential to seizure generation [7,37]. Assigning every resected channel an EZ label can consequently enlarge the positive set. Outcome supplies an additional validation dimension: a retrospective study found that the fraction of the observed SOZ resected did not closely track surgical outcome [39]. The treatment map, observed seizure onset, and outcome should therefore remain distinguishable quantities (Table 9).

This is structured uncertainty arising from the clinical definition, implantation and intervention. An algorithm that concentrates scores on a smaller subset can lose recall against a broad resection label, while an algorithm that follows that label closely can learn aspects of local treatment practice. Retaining the original observations makes such disagreement available for clinical adjudication. The issue also begins before annotation: human microelectrode recordings distinguish intensely recruited seizure cores from surrounding tissue displaying large electrographic signals [8], and a systematic review found limited evidence establishing the reliability of visual SOZ delineation [40]. A contact's recorded signal, assigned label and role in seizure generation are related stages of inference.

An actionable response is to preserve SOZ annotations, resection or ablation maps, electrode coordinates, follow-up outcomes and localization confidence as separate data fields. The Delphi framework of Ryvlin et al. provides a structured way to describe confidence in reported EZ localization [10]. These fields permit analyses of where a model agrees with onset interpretation, where it agrees with treatment, and whether either agreement relates to outcome. They also make disagreements available for adjudication instead of collapsing all clinical evidence into an apparently exact

channel target.

5.2 Making Patient-Level Evidence Comparable

The target definition and the evaluation unit jointly determine what a reported score means. Windows from the same channel, seizures from the same implantation, and channels from the same patient share substantial context. A split across those observations estimates a different use case from prediction for an unseen patient[32]. Weighting also changes the clinical question. Pooled channel metrics emphasize patients with more contacts; equal-patient metrics summarize performance for a typical included patient. These choices can change method rankings even when the underlying predictions remain fixed.

This distinction is particularly relevant when combining clinical sources. Quantitative iEEG reviews describe sensitivity to pathology, electrode placement and dataset construction[17,19]. A 2026 ILAE meta-analysis of connectivity-based localization found pooled added value for outcome prediction while identifying sample size, reporting and access to individual patient data as priorities for translation[30]. The field therefore needs a reproducible mapping from clinical records to the evaluated task: who was included, which channel was counted once, how its target was defined, and which patients informed preprocessing, fitting and threshold selection.

Our evaluation makes the patient this unit of separation and aggregation. The source-held-out analysis then asks a further question about transfer to a different recording population. The interictal analysis illustrates why cohort construction belongs beside the performance table: the original Omni-iEEG benchmark already uses subject-level separation, while our filtering and consolidation produce a different set of unique patient–channel pairs[24]. Its results describe that explicitly defined population. Agreement on a dataset name alone does not align inclusion rules, label semantics or endpoints.

Shared data formats such as iEEG-BIDS support this process[31]. Reproducible comparisons additionally require split identifiers, channel eligibility rules, preprocessing and threshold procedures, and predictions from the reported runs. This follows the emphasis of TRIPOD+AI on transparent specification of model development and evaluation[41]. With these elements fixed, a PR comparison estimates the effect of adding patient context to a particular predictor, and the component ablations identify the contribution of the model's reference operations.

5.3 Patient Context as a Modeling Variable

PR addresses a specific part of this inference problem: the reference against which a measured channel is interpreted. Anatomical normative maps show the value of comparing recordings with an explicit distribution of expected physiology[21,22]. PR uses another source of context, the distribution of the patient's observable channels. Its coordinate transformation allows a shared predictor to operate on the channel's position within that distribution. For a linear readout, Eq. (7) makes the consequence explicit: the same learned coefficients induce patient-dependent coordinate weights and an offset.

The experiments separate the reasons to use this information. Physiological associations identify measurements related to the channel annotations within patients. The half-channel CRPS experiment then tests the reference independently of classification: statistics estimated from one set of channels predict other channels from the same patient. Paired baseline comparisons test whether those relative coordinates help identify the annotated target. This progression connects a distributional property of the recordings to a modeling choice and then to localization performance.

DREChanNet combines this cross-channel context with temporal self-reference. Pre-onset subtraction expresses a channel's change from its own earlier activity, and cross-seizure aggregation retains both its average response and variability. The physiological and raw branches provide explicit measurements and learned waveform representations within this common design. Their joint performance and the PR ablations support using both forms of evidence. The interictal readout applies the same reference principle to separately trained representations, establishing a second application without tying PR to an ictal input sequence.

The reference remains conditional on the implanted set. SEEG implantation is guided by a presurgical hypothesis[6,7]; changing coverage or the mixture of abnormal and relatively unaffected contacts changes its distribution. PR consequently describes relative position in the observed system. Anatomical normative maps describe departure from an external physiological reference. Combining these two forms of context is a direct extension: anatomy can define the comparison group, while patient statistics can describe its recording-specific location and scale. Spatial perturbation methods for assessing implantation adequacy provide a relevant basis for evaluating such extensions[42].

5.4 Connecting Localization to Clinical Decisions

A channel score acquires clinical value through the decision it informs. The ranking experiment provides one concrete use: bringing relevant contacts forward for detailed review. In the displayed frozen run, DREChanNet achieves a Top-5 hit rate of 90.00%, with 63.00% precision and 21.98% recall at five channels (Fig. 4). These endpoints describe both the concentration of labeled EZ channels and the extent of the target covered by a short list. Their joint interpretation connects model output to the amount of evidence a clinician would inspect.

The other experiments address how that evidence becomes available. Additional seizures improve balanced classification in the fixed seizure-sufficiency cohort, supporting aggregation of repeated observations of the same implanted system. The interictal application supplies a complementary route when the available recording state differs. Clinical iEEG interpretation already integrates ictal and interictal observations[7]; a reusable reference operation can help organize these observations within their respective models. The source-held-out results further evaluate whether the resulting predictor transfers across the contributing populations.

The next assessment should measure how this channel evidence changes clinical work. A prospective study can present a fixed model's ranked contacts and corresponding physiological context during multidisciplinary review, then record review time, changes in the localization hypothesis, requests for additional evidence and agreement among assessors. Seizure outcome and preservation of function would assess the downstream value of the resulting decisions. DECIDE-AI provides a framework for examining both system performance and the human factors of this early clinical use[43]. Including

different pathologies, implantation strategies and postoperative outcomes would connect the defined channel task to the broader surgical population. This places the contribution of DREChanNet and PR at a useful point in the clinical chain: organizing patient-specific electrophysiological evidence so that its interpretation and consequences can be evaluated explicitly.

6 Conclusion

DREChanNet combines physiological descriptors and learned temporal representations for ictal channel localization, with PR processing linking both branches to the patient's observable channel distribution. The accompanying plug-in adds the same form of context to existing predictors and extends to an interictal readout. Feature associations, held-out-channel reference prediction, paired comparisons and component ablations support this design. By defining the patient-level task alongside the reference operation, the study connects a practical modeling contribution to the broader challenge of translating heterogeneous iEEG observations into interpretable localization evidence.

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Supporting Information

DREChanNet: Patient-Relative iEEG Model for Channel-Wise Epileptogenic Zone Localization

Supplementary methods and implementation settings for the accompanying manuscript.

S1 | Data preprocessing and patient separation

The ictal evaluation contains 80 patients, 256 seizure records, and 7,635 valid channels: 1,743 EZ and 5,892 NEZ channels. All recordings, windows, and channels belonging to one patient share the same role. The five outer folds contain 16, 16, 17, 15, and 16 patients, with 1,631, 1,498, 1,746, 1,425, and 1,335 channels, respectively. Each fold reserves 13 development patients for validation; the remaining 51, 51, 50, 52, and 51 patients form FIT. FIT determines preprocessing and learned parameters. Validation determines checkpoints and decision thresholds; the outer fold supplies the reported held-out predictions.

Onset-centered recordings and channel alignment. Ictal processing starts from the common onset-centered waveform cache. Each record occupies a 60-s frame at 250 Hz, corresponding to 15,000 samples per channel, with onset at the frame midpoint. Record identity combines the patient, recording/run, sample, and seizure identifiers. Channel names are canonicalized before matching the waveform and feature views. Matching uses these explicit identities and channel names, so the raw and physiological branches describe the same contact and seizure.
The cache carries the start and length of the genuinely observed signal inside each frame. Storage padding is kept separate from this observed interval. The inherited waveform samples are reused for all comparisons; downstream processing preserves the source channel montage and applies the window and normalization steps below. The spectral range specified for physiological features refers to retained PSD bins.

Window construction and validity. Each observation is a 2-s, 500-sample window centered at its stored onset-relative time t. Valid centers span approximately -29 to $+29$ s. The starting index in the 60-s frame is $\mathrm{round}[250(30 + t)] - 250$, and the endpoint is 500 samples later. A window is used only when the entire slice lies within the genuinely observed signal and the channel is available. This rule excludes windows that intersect padding or extend outside the recording. Nonfinite signal values fail the input check. The resulting fixed support contains 1,471,965 valid channel-windows across all 80 patients.
Arrays are stored with explicit seizure, channel, window, and sample axes and a corresponding validity mask. Every temporal mean and SD uses this mask. A missing channel/seizure observation contributes no value or count; a zero storage entry is not included as a measured waveform. Window times, valid sample bounds, and canonical channel order are checked against the source records before feature construction and model input.

Waveform conditioning for raw encoders. For each valid window x, the raw-model input is

$$x^{\mathrm{CNN}} = \mathrm{clip}\left(\frac{x - \mathrm{mean}(x)}{\max[\mathrm{SD}(x), 10^{-6}]}, -8, 8\right), \tag{S1}$$

where the mean and population SD are computed over that window's 500 samples. The same conditioning is applied to all raw baselines and the DREChanNet raw encoder. It requires no fitted population statistics. Cross-channel PR is applied later, after the window embeddings have been aggregated into channel representations.

Physiological input. The six measurements are beta power (13–30 Hz), low-gamma power (30–80 Hz), partial high-gamma power (80–125 Hz), RMS, line length, and spectral entropy (10–125 Hz). They are computed from the cached waveform before the CNN-specific amplitude standardization above, preserving amplitude information for RMS and line length. Computation retains the original float32 arithmetic.
Welch estimation uses a periodic Hann window, a 500-sample segment, an overlap parameter of 250 samples, density scaling, and no detrending. For a 500-sample input this gives a single full-length segment, with 0.5-Hz frequency bins. Bins from 10 Hz to the 125-Hz Nyquist limit are retained. Band integration uses the trapezoidal rule, followed by $\log(1 + P)$; beta and low gamma have exclusive upper boundaries, and the available high-gamma support includes 125 Hz.
RMS is $\sqrt{\mathrm{mean}(x^2) + 10^{-8}}$. Line length is the sum of absolute successive sample differences divided by 2 s. For entropy, PSD values are divided by their sum clipped below at 10^{-8}; the implementation evaluates $-\sum_m \pi_m \log(\pi_m + 10^{-8}) / \log(M + 10^{-8})$, where M is the number of retained bins. These numerical rules are shared by all models receiving the physiological representation.
For each seizure and channel, the mean measurement in valid pre-onset windows is subtracted from every valid window measurement. The resulting values are averaged over all valid windows in that seizure record. The mean and population SD across available seizures give two coordinates per measurement, hence 12 coordinates. Seven single-seizure patients remain included, with cross-seizure SD set to zero. All feature baselines receive these same 12 coordinates. DREChanNet retains a fixed 14-slot feature input whose two removed variance coordinates are identically zero.

Population preprocessing and patient references. Imputation and population scaling are estimated separately within each fold from FIT patients. A mean imputer supplies missing feature values, and models requiring population standardization use FIT means and SDs. For feature PR, unlabeled channels of the current patient first define coordinate-wise means and population SDs; the FIT-fitted imputer and any final standardizer are applied afterward. The same fitted objects are then used unchanged on validation and outer patients. This ordering keeps the patient's reference local while population preprocessing follows the training population.

S2 | Feature baseline configurations

Table S1 specifies the four feature classifiers displayed in the main comparison. Their model/fold configurations are inherited from the existing validation-selected settings and held identical between the original and PR conditions. PR replaces the 12 input coordinates by their patient-standardized counterparts. Linear SVM and MLP use the FIT-fitted population standardizer described in Section S1; random forest and XGBoost use the imputed inputs without this final scaling.

Training weights. Let C_p be the number of FIT channels of patient p, and N_y the number of FIT channels of class y. Estimator sample weights are proportional to $1/(C_p N_y)$, combining inverse patient-channel count and inverse FIT class frequency. Counts are recomputed inside each legal training population. These sample weights are supplied to all four estimators.

SVM calibration. Three patient-disjoint folds within FIT generate an out-of-fold decision margin for each FIT channel. Each subfit estimates its own imputer, standardizer, and weights from its sub-FIT patients. A one-dimensional logistic calibrator is fitted to these margins using inverse patient-size weights, without the additional inverse-class factor. Its settings are $C = 10^6$, LBFGS, maximum 1,000 iterations, and tolerance 10^{-8}. The final SVC is then fitted on all FIT patients. Validation and outer patients enter neither SVC fitting nor calibrator fitting. The feature implementation uses scikit-learn 1.7.2 and XGBoost 3.0.5.

S3 | Raw baselines and the final ictal model

S3.1 | Channel-level raw baseline adaptations

The raw baselines are the channel-level adaptations in Table S2. They share the input windows, masks, raw preprocessing, and patient roles described in Section S1. For a window embedding of dimension d, valid windows are averaged within each seizure/channel. The available-seizure mean and population SD are then concatenated coordinate-wise to form a $2d$-dimensional channel representation. These baseline representations use the full valid recording window sequence, without DREChanNet's pre-onset subtraction or physiological inputs. SEEGformer retains its cross-channel attention and masks invalid channels.

TimeConv-CNN's ictal ResNet is initialized from scratch. EEGNet's spatial kernel spans the single input channel; spatial and classifier weights are constrained to maximum norms 1 and 0.25, respectively. Its batch-normalization epsilon and momentum are 0.001 and 0.01. SEEG-Net aligns the temporal branches by truncating them to their common length; the squeeze-and-excitation reduction is $96 \to 12 \to 96$. SEEGformer projects each frequency branch to 64 coordinates, preserves masked channel attention, and applies branch heads to the corresponding 128-dimensional mean/SD channel block. Its heads are LayerNorm, Linear$(128, 32)$, GELU, dropout 0.2, and Linear$(32, 1)$. Including their channel classifiers, the original models contain 11,336,737, 2,329, 279,950, and 310,278 parameters, respectively, in the order of Table S2.

Original raw baselines minimize FIT-weighted binary cross-entropy with NEZ encoded as the positive internal class. The NEZ weight is $N_{\mathrm{EZ,FIT}}/N_{\mathrm{NEZ,FIT}}$. Each optimizer step uses one complete patient. AdamW has weight decay 10^{-4} and gradient-norm clipping at 1. The epoch limit is 30, with a six-epoch minimum and stopping patience of six epochs. Backbone training uses BF16, FP32 reductions, and TF32 disabled; validation and outer inference use FP32. Exact gradient caching partitions window encoding while retaining complete-patient pooling and gradients. Replay preserves random-number states and batch-normalization buffers. Thus chunk size does not define the patient reference or an independent training sample.

S3.2 | Frozen linear PR residuals in the main comparison

For each original raw baseline, the selected encoder and classifier are fixed in evaluation mode. If $a_{0,pc}$ is its EZ logit and h_{pc} its $2d$-dimensional channel representation, the augmented predictor is

$$a_{pc} = a_{0,pc} + \beta^{\mathsf{T}} R_p(h_{pc}) + b. \tag{S2}$$

Both β and b start at zero; epoch zero is an eligible identity correction. This adds 257, 481, 513, and 385 trainable parameters to TimeConv-CNN, EEGNet, SEEG-Net, and SEEGformer, respectively. No encoder or original classifier parameter is updated. Patient PR uses double-precision population moments over all valid channels. Its denominator is the SD when it exceeds 10^{-8} and 1 otherwise; a constant coordinate therefore maps to zero.

The residual uses the original FIT-weighted binary cross-entropy, averaged within each patient and then equally across FIT patients. Training is CPU FP32 AdamW, learning rate and weight decay 10^{-3}, gradient clipping at 1, and one complete FIT-population update per epoch. The maximum is 200 epochs, the minimum 10, and stopping patience 20. Validation selects the correction and its threshold. These frozen linear residuals generate the main raw-baseline PR results and the TimeConv PR ranking results. The TimeConv seizure-sufficiency comparison uses its separately trained, end-to-end PR classifier, with PR between channel aggregation and classification.

S3.3 | DREChanNet implementation

The final model has 9,329 parameters: 2,160 in the raw window encoder, 7,104 in the fused hidden layer, and 65 in the binary output layer. The encoder consists of Conv1D$(1, 12, 9$, stride 4), GELU, Conv1D$(12, 24, 7$, stride 4), GELU, and global average pooling. Its 24-dimensional window embeddings undergo same-channel pre-onset subtraction. Two stages, $[0, 8)$ and $[8, 20)$ s, each contribute cross-seizure mean and population SD, giving 96 raw coordinates. A channel/seizure contributes only where its stage contains valid windows.

The 96 raw coordinates and the 14-slot physiological vector enter a shared 64-unit GELU hidden layer through separate input blocks. Their fixed multipliers are $\sqrt{110/192}$ and $\sqrt{110/28}$; the output is one NEZ logit. Raw PR uses the rule above. Feature PR uses finite-value patient moments and sets a coordinate to zero

TABLE S1 | Feature baseline settings used for the main comparison. Vectors are ordered by outer folds 1–5. All inputs have 12 active coordinates; settings are shared between each original/PR pair.

Model	Fixed estimator settings	Fold-specific settings
Linear SVM	Linear-kernel SVC; probability fitting disabled; 512-MB kernel cache. FIT-only, threefold patient-level out-of-fold Platt calibration.	$C = (1, 1, 10, 1, 1)$.
Random forest	300 bootstrap trees; $\sqrt{d}$ features per split; weighted training; two CPU threads.	Maximum depth (none, 8, 8, none, none); minimum leaf size (5, 1, 5, 5, 1).
XGBoost	Binary logistic objective; histogram trees; learning rate 0.05; minimum child weight 5; row and column sampling 1; L_2 penalty 1; two CPU threads.	Maximum depth $(2, 2, 2, 4, 4)$; tree count $(500, 200, 200, 200, 500)$.
MLP	One 32-unit ReLU hidden layer; Adam; learning rate 10^{-3}; batch size 128; maximum 400 iterations; tolerance 10^{-5}; no internal early-stopping split.	L_2 coefficient $(10^{-4}, 10^{-2}, 10^{-2}, 10^{-2}, 10^{-2})$; convergence patience 20.

TABLE S2 | Ictal raw baseline adaptations. d is the window embedding dimension; the channel representation has $2d$ coordinates. Learning rates and chunk sizes apply to original backbone training. A chunk is a forward/replay partition within a complete-patient update.

Model	Implemented architecture	d	Channel dim.	Learning rate	Chunk
TimeConv-CNN	Hann STFT: FFT 256, hop 20; $\log(1 + \lvert S \rvert)$; interpolation to 224 frequency bins. Temporal convolutions $1 \to 16 \to 32$ with kernels $(3, 5)$ and $(3, 2)$, strides $(1, 5)$ and $(1, 2)$, and ReLU. ResNet-18 with 32 input maps; $512 \to 128$ projection.	128	256	3×10^{-4}	128
EEGNet	EEGNet-8,2 adaptation: 8 temporal filters of length 125; depthwise multiplier 2 and spatial kernel 1; separable temporal kernel 31; ELU; average pooling 4 then 8; dropout 0.5; flattened final map.	240	480	10^{-3}	1,024
SEEG-Net	Three 32-filter temporal branches: kernels $(63, 21, 3)$, strides $(5, 2, 1)$; batch normalization and ELU; 96-channel residual/SE block; dropout 0.5; bidirectional LSTM with 128 units/direction; attention pooling.	256	512	5×10^{-4}	256
SEEGformer	FFT 1,000; 0.5–125 Hz. Separate real, imaginary, and log-amplitude branches; each has width 64, two transformer blocks, four heads, feed-forward ratio 2, and dropout 0.2. Softmax-weighted branch logits.	192	384	5×10^{-4}	16

if its SD is at most 10^{-8} times its FIT-derived population scale. FIT-only imputation and scaling follow feature PR. The two zero feature slots remain zero. The raw-only and feature-only models have 8,433 and 1,025 parameters, respectively.

DREChanNet uses one complete patient per optimizer update, FP32 with TF32 disabled, AdamW at 3×10^{-4}, weight decay 10^{-3}, gradient clipping at 1, and the same 30/6/6 epoch limit/minimum/patience. The loss combines 0.75 FIT-weighted binary cross-entropy with 0.25 times one minus the patient-level soft macro-F1. The latter is the average soft F1 of EZ and NEZ, with a 10^{-8} denominator floor. Complete-patient encoding uses chunks of 4,096 windows. Removing pre-onset referencing removes the raw latent subtraction; the physiological input definition is retained. Removing either PR operation changes that branch's patient standardization, while the other model components follow the displayed ablation.

S4 | Selection, reference prediction, and evaluation

Checkpoint and threshold selection. For ictal neural models, validation prioritizes patient-mean macro-F1, then EZ-F1, then the earlier epoch. Feature baselines and raw residuals use EZ probabilities and call a channel positive at $q \geq \tau$. Their threshold grid is $0, 0.005, \ldots, 1$; ties prefer higher EZ-F1, the threshold closest to 0.5, and then the smaller threshold. DREChanNet retains its native NEZ-probability threshold convention: an EZ call occurs below its selected NEZ threshold. The threshold orientation is retained when replaying each saved model. All selected checkpoints, preprocessing objects, calibrators, and thresholds are fixed before the corresponding outer predictions.

Unseen-channel reference prediction. The CRPS experiment uses the physiological features and its existing five-path assessment allocation, separately from the supervised outer-fold allocation. Each path assesses 16 patients; its legal TRAIN population contains 27, 27, 26, 28, or 28 patients. Every patient is assessed once. Training-derived imputation, mean, and population SD define one common coordinate system for all forecast methods. Global moments use all legal TRAIN channels; same-source moments use only legal TRAIN channels from the query patient's source. Population moments are channel-weighted.

For each patient and repetition $r = 0, \ldots, 99$, channels are sorted using a fixed, label-independent SHA256 scheme based on the patient key, repetition, and channel key. The first $\lfloor C_p/2 \rfloor$ channels form A and the remainder form B. A predicts B, then B predicts A. Each channel is evaluated once per repetition, always outside its reference set. The global and same-source forecasts are fixed across these partitions. Location-only combines the reference-half mean with global SD; scale-only combines global mean with reference-half SD; full PR uses both reference-half moments.

Each coordinate is scored as a Gaussian location–scale forecast in the same TRAIN units. All 12 coordinates, including constants, remain in the average. Zero SD uses the point-distribution score $\lvert x - \mu \rvert$; no variance floor is inserted into CRPS. Losses are averaged across coordinates, evaluated channels, and repetitions within the patient, then equally across the 80 patients. The Gaussian forecast specifies a distributional probe for the reference

TABLE S3 | Patient roles for leave-one-source-out evaluation.

Held-out source	FIT	Validation	Outer
HUP	34	10	36
LZU	46	13	21
Multicenter	50	15	15
Pediatric	56	16	8

moments. It does not require the observed coordinates to follow a Gaussian distribution.

Patient endpoints and confidence intervals. For each patient, macro-F1 is the arithmetic mean of the EZ and NEZ F1 scores. Accuracy, AUROC, AP, class-specific F1, and ranking endpoints are also computed within patients. Multi-run results average each patient's endpoint across runs with different random initializations, then average patients equally. The main-model and ablation summaries follow this procedure. The ranking figure uses one frozen run per model.

Scores are sorted without rounding, with channel identifiers supplying deterministic tie order. AUROC gives half credit to tied positive/negative scores; AP groups equal scores at a common cutoff. MRR is the inverse rank of the first EZ channel. At K reviewed channels, precision is $E_p(K)/K$, recall is $E_p(K)/N_{p,\mathrm{EZ}}$, and hit rate is $\mathbf{1}[E_p(K) > 0]$. Fixed K is capped at the patient's available channel count. A fractional budget q uses $\lceil qC_p \rceil$ channels. A hit-rate percentage is consequently distinct from the percentage of EZ channels recovered.

CRPS and displayed ranking intervals use 10,000 paired patient-bootstrap resamples from a fixed pseudorandom generator, retaining the original counts of 36 HUP, 21 LZU, 15 multicenter, and eight pediatric patients by resampling within source. Each resample is shared across compared methods; percentile endpoints give nominal 95% intervals. Cross-channel repeats and seizure subsets are collapsed within patients before resampling. Feature-association intervals use the feature analysis's 10,000 patient-bootstrap resamples; its 12 two-sided Wilcoxon tests use Benjamini–Hochberg correction. The proportion of positive effects is the fraction of all 80 patients whose signed rank-biserial effect exceeds zero, with ties displayed separately.

Source transfer and seizure sufficiency. The leave-one-source-out roles are given in Table S3. The entire test source is absent from FIT and validation; its unlabeled channels provide its patient PR moments. The linear SVM uses $C = 1$ and FIT-only threefold patient-level Platt calibration in this analysis. For each displayed method, endpoints are averaged across the reported runs within source. Center Mean is the arithmetic mean of the four source means; Worst is their minimum.

Seizure sufficiency retains 57 patients with at least three seizures and 5,735 channels common to all evaluated conditions. One-, two-, and all-seizure inputs use fixed checkpoints and thresholds. Every eligible seizure subset is evaluated when there are at most 50 combinations; otherwise 50 fixed subsets are sampled. Endpoints are averaged across subsets within patient. The case illustration is a retrospectively selected 49-channel example from a separately evaluated 11,425-parameter development encoder. Its displayed channel decisions use that encoder's frozen thresholds; the main cohort and ablation results use the final 9,329-parameter model.

S5 | Patient-based interictal PR application

Eligibility and canonical channels. The evaluated Omni-iEEG subset has 124 patients, 137 EDF records, and 10,202 unique labeled patient–channel pairs: 1,113 pathological and 9,089 normal. Its source patient counts are Open-iEEG 103, HUP 10, and SourceSink 11; outer-fold sizes are 25, 24, 25, 25, and 25. Patients must have both labeled classes after filtering and channel consolidation. The target is constructed in the following priority order:

1. Exclude channels not marked good.
2. Assign normal to a good, nonresected channel from a seizure-free patient.
3. Otherwise assign pathological to a good channel annotated as SOZ.
4. Keep remaining good channels as unlabeled reference context; exclude them from loss and endpoints.

Repeated records of a canonical channel within a patient are consolidated into one evaluation unit. Thus eligibility, label availability, and record consolidation determine the reported channel count. The same canonical good-channel set supplies the label-free PR reference, including channels without evaluation labels.

Frozen backbone and aggregation. The historical fold-specific TimeConv backbone is fixed throughout readout training. It uses the pinned official spectral transform: Morlet-like spectra over 10–300 Hz with 224 frequency segments, followed by the official image normalization. The official NeuralCNN produces a 32-dimensional embedding before its original fully connected/normalization classifier. This is distinct from the 128-dimensional ictal adaptation in Table S2.

The baseline backbone uses ImageNet ResNet-18 initialization and is fitted on the four outer FIT folds for 30 fixed epochs, with AdamW learning rate 10^{-4}, weight decay 10^{-3}, and gradient clipping at 1. At epoch k, training uses the kth nonoverlapping 2-s segment; the 30-epoch schedule traverses the 60-s record. Its epoch-30 checkpoint supplies the frozen representation for the reported incremental readout.

Interictal preprocessing applies a 60-Hz notch with 2-Hz notch width using MNE, then resamples each channel to 1 kHz when required. A fixed 60-s interval beginning 1 s after recording onset is retained. This interval is divided into 30 nonoverlapping 2-s segments. For a canonical channel, repeated-record embeddings and logits are first averaged within each segment; segment-level values are then averaged. The frozen baseline probability is the sigmoid of the mean logit. Channel batches of 16 retain the historical feature-extraction behavior.

Incremental PR readout. For the patient embedding matrix h_p, use

$$\widetilde{h}_{pcj} = \frac{h_{pcj} - \mu_{pj}}{\sqrt{v_{pj} + 10^{-5}}}, \qquad a_{pc} = a_{0,pc} + \beta^{\mathsf{T}} \widetilde{h}_{pc} + b_{g(p)}, \tag{S3}$$

where v_{pj} is population variance across good canonical channels and $g(p)$ identifies one of the three sources. The 32 coefficients and three source biases give 35 new parameters, initialized to zero. No source-specific direction matrix or additional nonlinear layer is used for this reported PR readout.

CPU FP32 AdamW uses learning rate 10^{-3}, weight decay 10^{-4}, gradient clipping at 1, and one patient update at a time. Pathological and normal weights are 2 and 1 and are normalized within each patient. With P FIT patients and P_g from source g, the explicit regularizer is

$$\mathcal{R} = 10^{-3}\left[\frac{1}{32}\sum_{j=1}^{32}\beta_j^2 + \frac{1}{3}\sum_{g=1}^{3}\frac{P}{3P_g}b_g^2\right]. \tag{S4}$$

Thus the shared coefficients use a mean-square penalty, and source-bias penalties are weighted inversely by FIT source frequency.

Within each outer FIT population, three source-stratified patient folds select the readout training length from 50 epochs using mean AUROC, then AP, then the earlier epoch. These folds assess the readout on representations from the fixed outer-FIT backbone. The selected epoch's out-of-fold readout predictions determine a threshold on the 0.005 probability grid, prioritizing patient-mean macro-F1 and then pathological F1. The readout is reinitialized and fitted on all outer FIT patients for the selected number of epochs. The baseline has its own FIT-derived threshold. Outer patients are excluded from both backbone and readout fitting.